Software Engineering Interview Guide

Q1. What is SDLC?

SDLC stands for Software Development Life Cycle. It is a structured process used to develop, test, deploy, and maintain software.

Example:
Suppose a company wants to build a food-delivery application.

Q2. What is the difference between Agile and Waterfall?

Waterfall follows a mostly sequential approach where one phase is completed before moving to the next. **Agile** develops software incrementally and iteratively, with frequent feedback and adaptation.

Example:

Q3. What is Scrum?

Scrum is an Agile framework used to organize and manage software development work in short, iterative cycles called sprints.

Example:

Q4. What is a sprint?

A sprint is a fixed-length period during which a Scrum team works toward a specific goal and produces a usable increment of the product.

Q5. What is Git?

Git is a distributed version control system used to track changes in source code and collaborate with other developers.

Q6. What is the difference between Git and GitHub?

Git is the version control system. **GitHub** is a platform/service that hosts Git repositories and provides collaboration features.

Q7. What is a branch in Git?

A branch is an independent line of development in a Git repository. It allows you to work on a feature without directly changing the main branch.

Q8. What is a merge in Git?

Merge combines changes from one branch into another branch.

Q9. What is a Pull Request?

A Pull Request (PR) is a request to merge changes from one branch into another, usually accompanied by review and automated checks.

Q10. What is the difference between merge and rebase?

Merge combines histories and may create a merge commit, while **Rebase** moves feature commits on top of another branch for a linear history.

Q11. What is CI/CD?

CI/CD is a set of practices and automation used to build, test, and deliver software more reliably and frequently.

Q12. What is the difference between Unit Testing and Integration Testing?

Unit Testing tests small, isolated units of code, while **Integration Testing** tests whether multiple components work correctly together.

Q13. What is debugging?

Debugging is the process of finding, understanding, and fixing errors or unexpected behavior in a program.

Q14. What is code review?

Code review is the process where other developers examine code changes before they are integrated into the main codebase.